Compass Final: one clean test, at Robinhood’s fees, of every rule the final report left standing

Repository: georgeredo, branch `claude/lucid-heisenberg-bpyq7k`. **Prereg:** `PREREG_COMPASS_FINAL.md`, committed alone with its HOLDOUT_LEDGER row in `4bc546f` before any fetch or computation. **Fetch and run code:** `f6d2889` (before T1–T6 were run). **Built:** 2026-10-24 by `report_final.py`; every figure below is read from `out/results.json` and the trade files beside it.

In plain English

The owner asked for the research to be reverse-engineered “until you figure out a final that works”. This study did it the honest way: it picked the survivors from the final report by a rule fixed in advance, and gave each one test, run once, on market data no study had read for that rule, at Robinhood’s fees, with nothing tuned.

Nothing passed. Seven tests ran, and all seven fail the pre-registered rule. The headline for each, at Robinhood’s fees under the pre-registered settlement:

- **T1 — SPY/QQQ iron butterfly ±5 at 15:00 and 15:30 (fly5 iron; hyp C §6):** 1,021 trades, mean -3.32 % of max risk a trade, 95 % interval [-8.95, +2.83], win rate 48 %. **FAIL** (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year).
- **T2 — SPY 2-wide iron condor at ± the ATM straddle, 11:30–13:30 (condor_em1):** 708 trades, mean -4.38 % of max risk a trade, 95 % interval [-16.19, +9.21], win rate 82 %. **FAIL** (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year).
- **T3 — SPY 2-wide 20-delta iron condor, 13:00–15:00 (condor_d20):** 799 trades, mean +0.21 % of max risk a trade, 95 % interval [-9.19, +9.79], win rate 81 %. **FAIL** (Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year).
- **T4 — SPY 2-wide 25-delta credit spread at the boundary touch (hyp B touch pick):** 1,010 trades, mean -4.80 % of max risk a trade, 95 % interval [-13.11, +2.63], win rate 83 %. **FAIL** (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year).
- **T5 — SPXW 13:00 ATM straddle to settlement, ex FOMC/CPI/NFP (T-E, H13):** 246 trades, mean -1.16 % of credit a trade, 95 % interval [-12.26, +12.14], win rate 59 %. **FAIL** (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≥ 300 trades, ≤ 1 losing year).
- **T6 — SPXW 14:05 strangle at ±0.5 EM to settlement, ex FOMC and month-end (H15):** 280 trades, mean +2.37 % of credit a trade, 95 % interval [-27.19, +30.73], win rate 73 %. **FAIL** (Bonferroni lower end > 0, top-5 ≤ 100 %, ≥ 300 trades, ≤ 1 losing year).
- **T7 — QQQ overnight reversal of the late-day move (Compass H1):** 249 trades, mean +0.082 % of notional a trade, 95 % interval [-0.007, +0.141], win rate 59 %. **FAIL** (Bonferroni lower end > 0, ≥ 300 trades).

What sank the SPY/QQQ defined-risk family is the after-hours exercise risk, measured as the owner fixed it. These structures are held to the close on physically settled ETF options, so an option that is in the money at the close, or that the other side chooses to exercise after the close, turns into 100 shares that gap overnight. Priced the way AFTERHOURS.md priced it (case (b): our longs exercised on the official close, each short assigned if it is in the money at 17:30, the shares marked at the next 09:30 open), the late iron butterfly is **-3.32 %** of max risk a trade. Settled at intrinsic on the official close, which ignores that risk, the same trades read +2.18 % [+0.53, +3.29]: a small edge that exists only on paper. Closing at 15:58 to avoid the risk costs it: -0.90 % [-1.96, +0.05]. The condors and the touch spread show the same shape: they win most of the time (81–83 %) and their average losing trade is 119–144 % of max risk, more than the structure can lose at expiry, because an assignment leaves stock that moves overnight.

The SPXW lanes (T5, T6) could not pass on count: the fresh span holds 246 and 280 trades, under the 300 the owner requires (stated in the prereg before anything was read). Their returns also say no: T5 -1.16 % and T6 +2.37 % of the credit, with intervals about ±12 and ±29 points wide. One session, 2018-02-05, cost −$7,318 on T5 and −$6,078 on T6: years of credit gone in an afternoon.

H1 on QQQ (T7) is the closest thing to a positive: +0.082 % of notional a trade ($54 a 100-share trade) on 249 trades from 2025-10-13 to 2026-10-22, 95 % interval [-0.007, +0.141]. It still fails: under 300 trades (stated in advance) and its Bonferroni interval [-0.041, +0.175] crosses zero. It agrees with the earlier read (+0.038 %) without proving anything; its forward N at the earlier mean is 2,763 trades, about eleven years.

S17 index put-writing had no unread history (its test read all the option data Theta serves), so it goes to the forward test only. It is now in step 8 of the daily job (first roll 2026-11-20, no verdict before 147 closed cycles, about six years). H7 and QQQ's turn of the month were not candidates: they have 77–172 trades, under the 300 the selection rule requires. H7 is already forward-tracked.

The checks hold. 70 trades re-derived from the raw files by separate plain-loop code: 490 fields, **0 differences**. The lookahead check: **not run — usage ended** (started, stopped before it finished; no result read). Every run path was first checked on sessions the research had already read and reproduced the committed rows exactly (FETCH_LOG.md).

What it means. There is no rule here to trade. The power was low by construction (section 7: every MDE was two to five times the earlier means), so a FAIL is weak evidence that a small edge does not exist. But every family failed on more than its interval: the means are at or below zero, more than one year lost money, and the SPY/QQQ structures lose their paper edge to the after-hours exercise risk. No rulebook is written. The forward paper tests already in the daily job (fly5, condor_em1, condor_d20, H5/H6, H7) keep running to their own N, and S17 joins them.

The verdict, test by test

k = 7 tests; the Bonferroni interval is the two-sided 1 − 0.05/7 percentile interval of the 20-session moving-block bootstrap (audit_lib, 2,000 draws, seed 20261022). Robinhood fees; the primary settlement (case (b) for SPY/QQQ, Cboe's close for SPXW, the next 09:30 open for T7).

test	family	trades	sessions traded / in the test	mean a trade	95 % interval	Bonferroni interval	top-5 share of profit	years ≤ 0	verdict
T1	SPY/QQQ late ATM butterflies (rows 21a, 21c, 25, 26b)	1,021	415 / 479	**-3.32** % of max risk	[-8.95, +2.83]	[-11.62, +4.39]	n/a (total ≤ 0)	2012, 2014, 2015, 2018, 2019, 2021, 2022	**FAIL** (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year)
T2	SPY midday iron condors (rows 20b, 26a, 26c)	708	205 / 266	**-4.38** % of max risk	[-16.19, +9.21]	[-20.16, +13.21]	n/a (total ≤ 0)	2018, 2020, 2021, 2022	**FAIL** (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year)
T3	SPY midday iron condors (rows 20b, 26a, 26c)	799	212 / 266	**+0.21** % of max risk	[-9.19, +9.79]	[-13.08, +14.16]	1199 %	2018, 2021, 2022	**FAIL** (Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year)
T4	SPY boundary-touch credit spread (row 20a)	1,010	218 / 266	**-4.80** % of max risk	[-13.11, +2.63]	[-16.22, +4.98]	n/a (total ≤ 0)	2018, 2020, 2021	**FAIL** (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year)
T5	SPXW 13:00 straddle (row 6d)	246	246 / 297	**-1.16** % of credit	[-12.26, +12.14]	[-17.46, +16.97]	n/a (total ≤ 0)	2018, 2021, 2022	**FAIL** (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≥ 300 trades, ≤ 1 losing year)
T6	SPXW 14:05 strangle (row 6e)	280	280 / 297	**+2.37** % of credit	[-27.19, +30.73]	[-37.98, +39.20]	n/a (total ≤ 0)	2018, 2021	**FAIL** (Bonferroni lower end > 0, top-5 ≤ 100 %, ≥ 300 trades, ≤ 1 losing year)

test	family	trades	sessions traded / in the test	mean a trade	95 % interval	Bonferroni interval	top-5 share of profit	years ≤ 0	verdict
T7	Compass H1 on QQQ (row 32 QQQ)	249	249 / 259	**+0.082** % of notional	[-0.007, +0.141]	[-0.041, +0.175]	57 %	none	**FAIL** (Bonferroni lower end > 0, ≥ 300 trades)

Families with no unread data: **S17** (forward only, wired into the daily job). Not candidates: H7 and QQQ turn of the month (< 300 trades). No test passed, so no rulebook and no new forward N for a passing rule.

What was tested, and on what (stage 1 and the prereg, in short)

- **Selection rule (fixed first):** a page-1 row with a Robinhood-fee mean > 0, a block-20 lower end > 0 on its own period, ≥ 300 trades, and not vendor-recorded, a preview or a base-trade row. 13 rows passed, in 9 families (prereg §1).
- **Fresh data:** Theta serves SPY bars, SPY EOD and SPX prints from 2017-01-03 and QQQ from 2012-06-01, and SPY/QQQ after-hours bars throughout (stage-1 probes, `out/probe_entitlement*.json`). The fresh span ends 2022-05-12: from 2022-05-13 SPXW was read intraday on every session. Every session on which an earlier study held or read an index-product option inside the family's window (V8, S03, S16b, Compass Close's gate sample) is out; half days are out.
- **Sessions in the test:** SPY 266 (2017-01-04 → 2022-04-27), QQQ 339 (2012-06-01 → 2022-04-27), SPXW 297 (2017-01-03 → 2022-04-27), and QQQ stock 2025-10-13 → 2026-10-22 for T7. 8 QQQ sessions came back empty from Theta twice and carry no trade (FETCH_LOG.md).
- **Fees:** Robinhood (`docs/FEES_ROBINHOOD.md`) primary; each study's own fees as a line. **Fills** at the NBBO: buy at the ask, sell at the bid.
- **Settlement:** physically settled SPY/QQQ held to the close by AFTERHOURS.md's case (b) (the after-hours prices could be fetched, so the 15:59 close-out fallback was not used); SPXW on Cboe's official close; T7 at the next 09:30 open.

Power: the MDE committed before reading, and at the realised n

test	planned n (prereg)	realised trades	MDE per trade, prereg SD at realised n	MDE per session, prereg SD	MDE per trade, realised SD	realised SD a trade
T1	1,210	1,021	15.92	19.97	5.21	47.13
T2	1,330	708	17.00	24.37	13.25	99.81
T3	1,330	799	16.00	23.96	11.86	94.94
T4	1,601	1,010	14.23	23.63	9.12	82.09
T5	246	246	18.02	18.02	19.98	88.72
T6	281	280	29.11	29.11	49.26	233.38
T7	257	249	0.179	0.179	0.188	0.839

Units as each test's (percentage points of its unit). Every MDE is several times the effect the earlier reads found, as the prereg said it would be.

T1 — SPY/QQQ iron butterfly ±5 at 15:00 and 15:30 (fly5 iron; hyp C §6)

Verdict: FAIL (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year). Unit: % of max risk. Settlement: case (b). Every trade had its official close, decision price and next open (528 name-sessions; decision-price coverage 100 %).

trades	sessions traded / in test	win %	avg win %	avg loss %	median %	mean %	mean $	total $	profit factor	max drawdown $	longest drawdown (sessions)
1,021	415 / 479	47.6	+23.10	-27.32	-0.80	**-3.32**	-10.72	−$10,949	0.81	$15,319	227

Best trade $1,065 (+252.9 %, 2020-02-21 15:30, SPY iron); worst −$1,106 (-360.3 %, 2022-02-23 15:30, SPY iron).

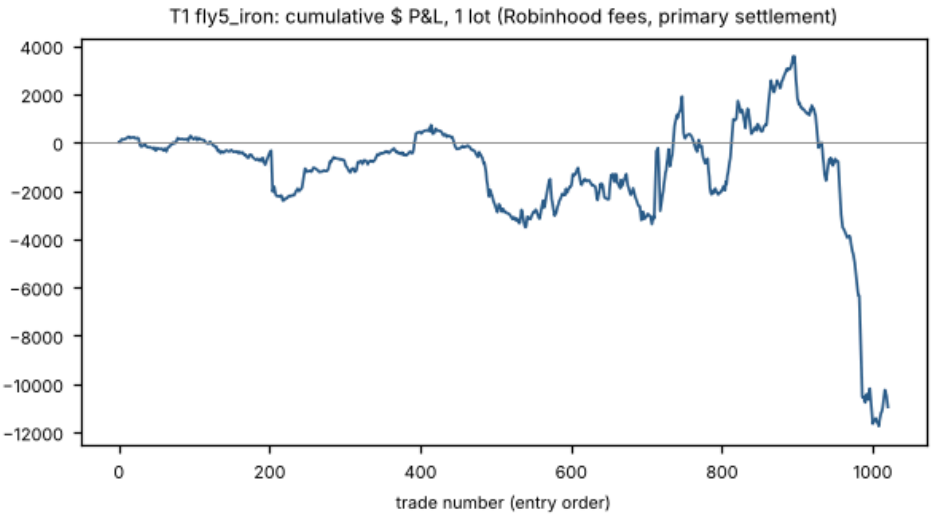
$1,000: at a fixed 1 lot it ends at −$9,949 (low −$10,746, first at or below $0 on 2015-08-21); at 10 % per trade (the final report's method B) it ends at $0, wiped out on 2022-02-23. A case (b) loss can exceed the structure's max risk (shares assigned and gapping overnight), so 10 % of the balance can lose more than 10 %.

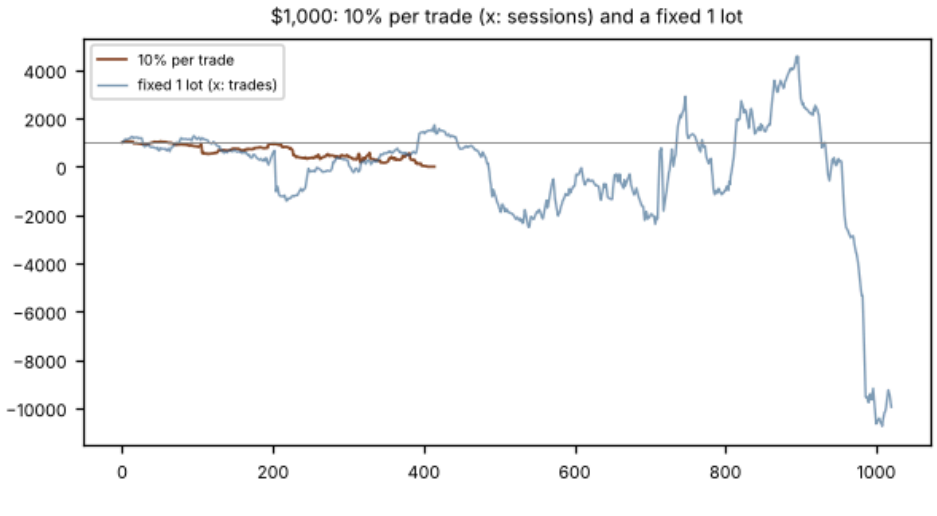
line (no verdict)	trades	mean	95 % interval	total $
primary, Robinhood fees	1,021	**-3.32**	[-8.95, +2.83]	−$10,949
(0) intrinsic at the official close [carries the exercise risk it does not price]	1,021	+2.18	[+0.53, +3.29]	$7,936
(a) exercise by exception on C	1,021	-2.67	[-8.68, +1.68]	−$9,725
(c) both sides decided at D	1,021	-3.32	[-8.95, +2.83]	−$10,949
the tracker's 15:58 exit (NBBO, Robinhood both sides)	1,021	-0.90	[-1.96, +0.05]	−$4,048
the study's own convention: (0) at the official close, $0.65 a leg	1,021	+1.63	[-0.01, +2.73]	$5,487
primary settlement at the study's own fees	1,021	-3.82	[-9.45, +2.33]	−$13,166
fly5 call (case (b), Robinhood)	1,027	-4.96	[-10.50, +1.38]	−$20,072
fly5 put (case (b), Robinhood)	1,026	-6.09	[-11.48, -0.23]	−$25,228
SPY alone	459	-2.17	[-9.33, +6.48]	−$2,703
QQQ alone	562	-4.26	[-9.17, +1.22]	−$8,246
sensitivity: Compass Close's P&L-sample sessions out	436	+0.11	[-2.55, +2.14]	$234
sensitivity: EOD-marked sessions out	0	—	—	every in-test session carries an EOD mark (stage 1): n = 0
sensitivity: sessions with	official - 15:59 bar	> $0.25 out	1,017	-3.50

by year	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
trades	29	50	72	85	96	151	120	99	118	147	54
mean %	-0.77	+1.23	-1.55	-4.43	+2.68	+0.39	-1.74	-3.40	+8.76	-9.61	-43.22
total $	−$105	$294	−$527	−$1,672	$1,198	$192	−$701	−$1,642	$3,949	−$4,926	−$7,010

by weekday	Mon	Tue	Wed	Thu	Fri
trades	147	6	191	25	652
mean %	-5.10	-0.95	-8.18	+2.44	-1.74

by entry time	15:00	15:30
trades	515	506
mean %	-2.87	-3.77





T2 — SPY 2-wide iron condor at ± the ATM straddle, 11:30–13:30 (condor_em1)

Verdict: FAIL (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year). Unit: % of max risk. Settlement: case (b). Every trade had its official close, decision price and next open (205 name-sessions; decision-price coverage 100 %).

trades	sessions traded / in test	win %	avg win %	avg loss %	median %	mean %	mean $	total $	profit factor	max drawdown $	longest drawdown (sessions)
708	205 / 266	82.1	+26.03	-143.50	+12.25	**-4.38**	-6.32	−$4,477	0.85	$6,764	217

Best trade $645 (+408.1 %, 2020-12-18 12:00, SPY condor); worst −$1,147 (-775.0 %, 2022-02-23 11:30, SPY condor).

$1,000: at a fixed 1 lot it ends at −$3,477 (low −$5,228, first at or below $0 on 2018-02-07); at 10 % per trade (the final report's method B) it ends at $0, wiped out on 2020-04-06. A case (b) loss can exceed the structure's max risk (shares assigned and gapping overnight), so 10 % of the balance can lose more than 10 %.

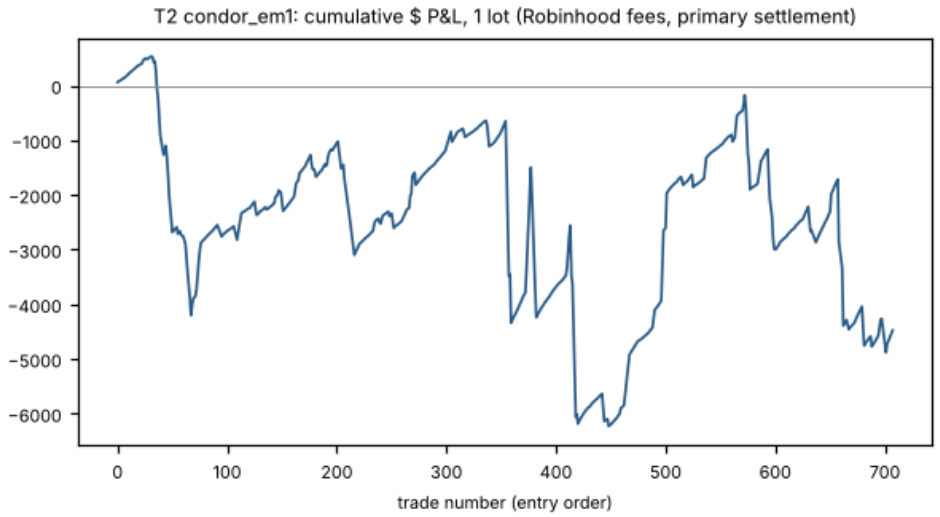
line (no verdict)	trades	mean	95 % interval	total $
primary, Robinhood fees	708	**-4.38**	[-16.19, +9.21]	−$4,477
(0) intrinsic at the official close [carries the exercise risk it does not price]	708	+1.11	[-6.21, +7.37]	$1,471
(a) exercise by exception on C	708	-5.70	[-16.62, +4.90]	−$6,068
(c) both sides decided at D	708	-3.55	[-15.28, +9.65]	−$3,463
the tracker's 15:58 exit (NBBO, Robinhood both sides)	708	-3.81	[-10.61, +2.50]	−$4,381
the study's own convention: (0) at the official close, $0.65 a leg	708	-0.27	[-7.61, +5.99]	−$228
primary settlement at the study's own fees	708	-5.72	[-17.51, +7.89]	−$6,123

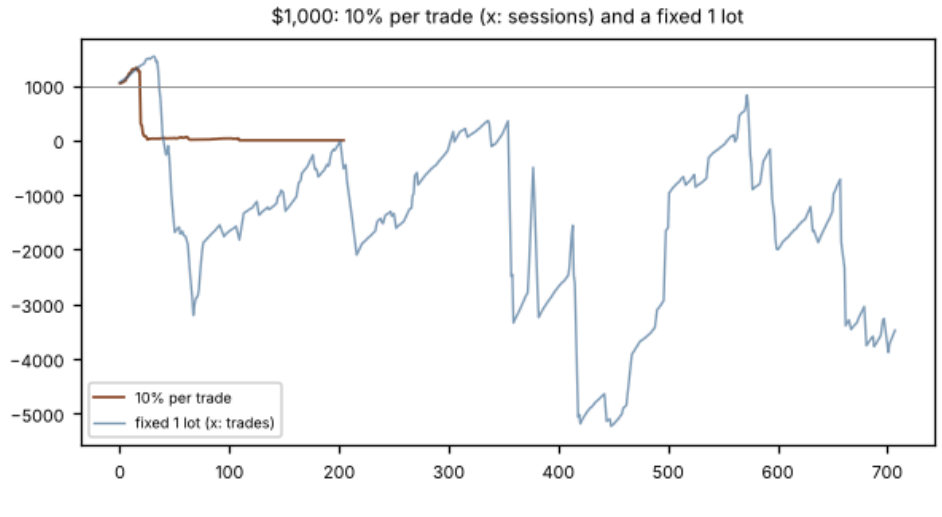
line (no verdict)	trades	mean	95 % interval	total $
sensitivity: Compass Close's P&L-sample sessions out	29	+9.21	—	$494
sensitivity: EOD-marked sessions out	0	—	—	every in-test session carries an EOD mark (stage 1): n = 0
sensitivity: sessions with	official - 15:59 bar	> $0.25 out	699	-3.37

by year	2017	2018	2019	2020	2021	2022
trades	35	187	115	166	135	70
mean %	+6.90	-11.21	+11.20	-5.64	-3.96	-15.18
total $	$446	−$3,331	$2,243	−$1,297	−$928	−$1,610

by weekday	Mon	Wed	Thu	Fri
trades	200	262	10	236
mean %	-22.73	-14.29	-5.96	+22.23

by entry time	11:30	12:00	12:30	13:00	13:30
trades	158	157	138	129	126
mean %	-6.05	+1.30	-8.14	-1.78	-7.89





T3 — SPY 2-wide 20-delta iron condor, 13:00–15:00 (condor_d20)

Verdict: FAIL (Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year). Unit: % of max risk. Settlement: case (b). Every trade had its official close, decision price and next open (212 name-sessions; decision-price coverage 100 %).

trades	sessions traded / in test	win %	avg win %	avg loss %	median %	mean %	mean $	total $	profit factor	max drawdown $	longest drawdown (sessions)
799	212 / 266	81.4	+27.74	-119.87	+12.51	**+0.21**	+0.87	$691	1.02	$5,718	97

Best trade $642 (+398.6 %, 2020-12-18 13:30, SPY condor); worst −$1,149 (-766.0 %, 2022-02-23 14:00, SPY condor).

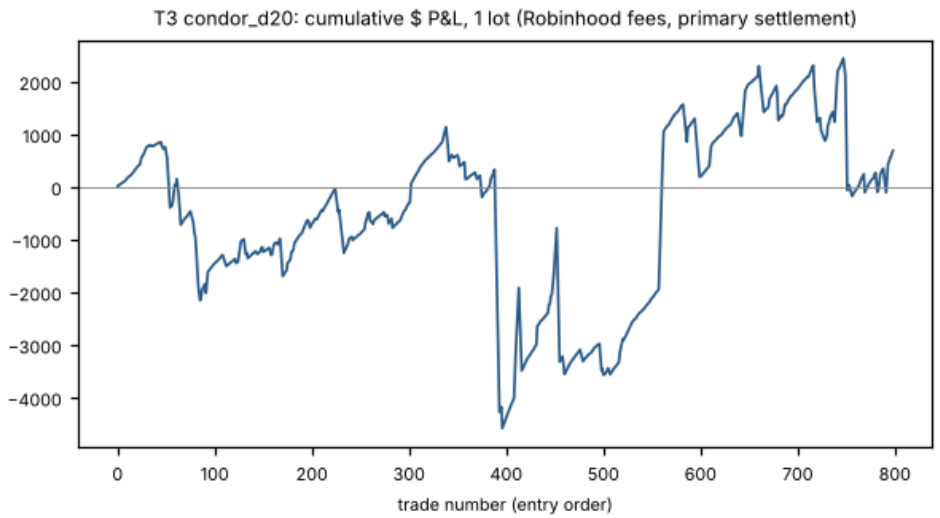
$1,000: at a fixed 1 lot it ends at $1,691 (low −$3,576, first at or below $0 on 2018-04-04); at 10 % per trade (the final report's method B) it ends at $0, wiped out on 2020-04-06. A case (b) loss can exceed the structure's max risk (shares assigned and gapping overnight), so 10 % of the balance can lose more than 10 %.

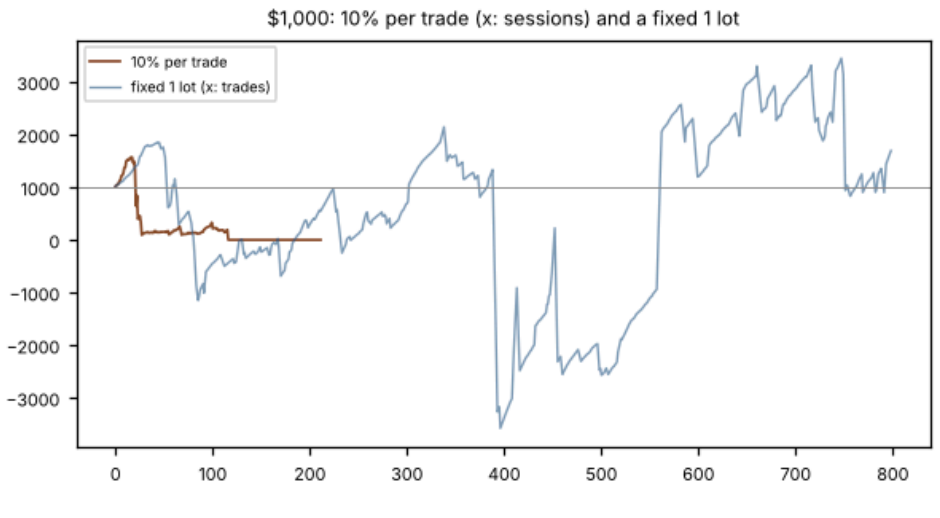
line (no verdict)	trades	mean	95 % interval	total $
primary, Robinhood fees	799	**+0.21**	[-9.19, +9.79]	$691
(0) intrinsic at the official close [carries the exercise risk it does not price]	799	+3.48	[-1.22, +7.36]	$4,659
(a) exercise by exception on C	799	+0.87	[-7.73, +8.51]	$1,778
(c) both sides decided at D	799	+0.29	[-9.44, +9.78]	$695
the tracker's 15:58 exit (NBBO, Robinhood both sides)	799	-2.09	[-6.60, +2.14]	−$2,750
the study's own convention: (0) at the official close, $0.65 a leg	799	+2.09	[-2.58, +5.96]	$2,742
primary settlement at the study's own fees	799	-1.13	[-10.53, +8.48]	−$1,157
sensitivity: Compass Close's P&L-sample sessions out	38	+11.25	—	$783
sensitivity: EOD-marked sessions out	0	—	—	every in-test session carries an EOD mark (stage 1): n = 0
sensitivity: sessions with	official - 15:59 bar	> $0.25 out	789	+1.08

by year	2017	2018	2019	2020	2021	2022
trades	52	187	130	200	160	70
mean %	+5.95	-4.97	+5.91	+2.15	-1.40	-2.62
total $	$564	−$1,554	$1,267	$932	−$328	−$188

by weekday	Mon	Wed	Thu	Fri
trades	220	289	18	272
mean %	-18.00	-2.75	-31.43	+20.19

by entry time	13:00	13:30	14:00	14:30	15:00
trades	181	170	158	150	140
mean %	-0.34	+0.46	-7.80	+5.20	+4.34





T4 — SPY 2-wide 25-delta credit spread at the boundary touch (hyp B touch pick)

Verdict: FAIL (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≤ 1 losing year). Unit: % of max risk. Settlement: case (b). Every trade had its official close, decision price and next open (218 name-sessions; decision-price coverage 100 %).

trades	sessions traded / in test	win %	avg win %	avg loss %	median %	mean %	mean $	total $	profit factor	max drawdown $	longest drawdown (sessions)
1,010	218 / 266	83.0	+18.56	-118.63	+8.05	**-4.80**	-8.01	−$8,088	0.78	$11,353	208

Best trade $626 (+353.6 %, 2020-12-18 15:57, SPY call spread); worst −$1,083 (-668.6 %, 2020-04-06 11:49, SPY call spread).

$1,000: at a fixed 1 lot it ends at −$7,088 (low −$9,445, first at or below $0 on 2018-02-02); at 10 % per trade (the final report's method B) it ends at $0, wiped out on 2018-02-02. A case (b) loss can exceed the structure's max risk (shares assigned and gapping overnight), so 10 % of the balance can lose more than 10 %.

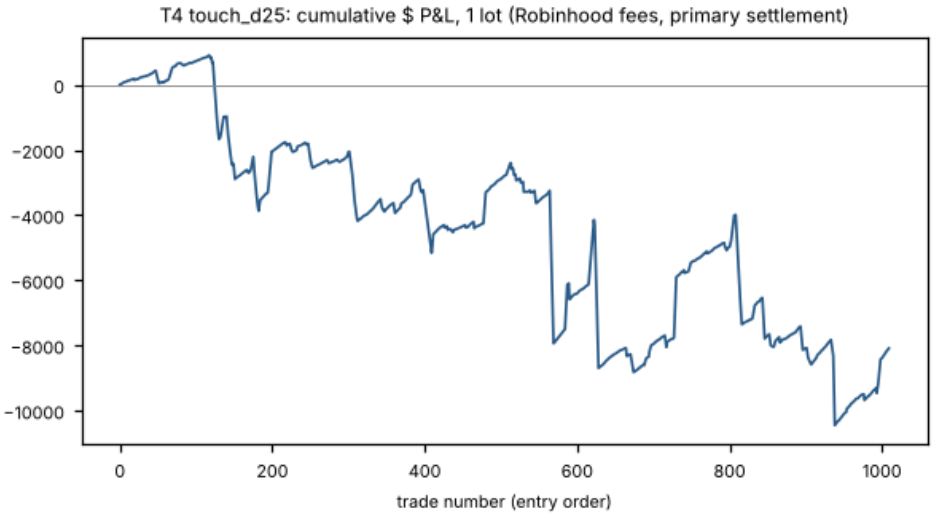
line (no verdict)	trades	mean	95 % interval	total $
primary, Robinhood fees	1,010	**-4.80**	[-13.11, +2.63]	−$8,088
(0) intrinsic at the official close [carries the exercise risk it does not price]	1,010	-0.39	[-4.22, +2.68]	−$640
(a) exercise by exception on C	1,010	-3.39	[-9.54, +2.25]	−$5,441
(c) both sides decided at D	1,010	-3.16	[-10.66, +3.96]	−$5,206
the study's own convention: the 16:00 bar's close, $0.65 a leg	1,010	-0.86	[-4.94, +2.30]	−$1,447
primary settlement at the study's own fees	1,010	-5.42	[-13.72, +2.01]	−$9,221
sensitivity: Compass Close's P&L-sample sessions out	92	+3.81	—	$665

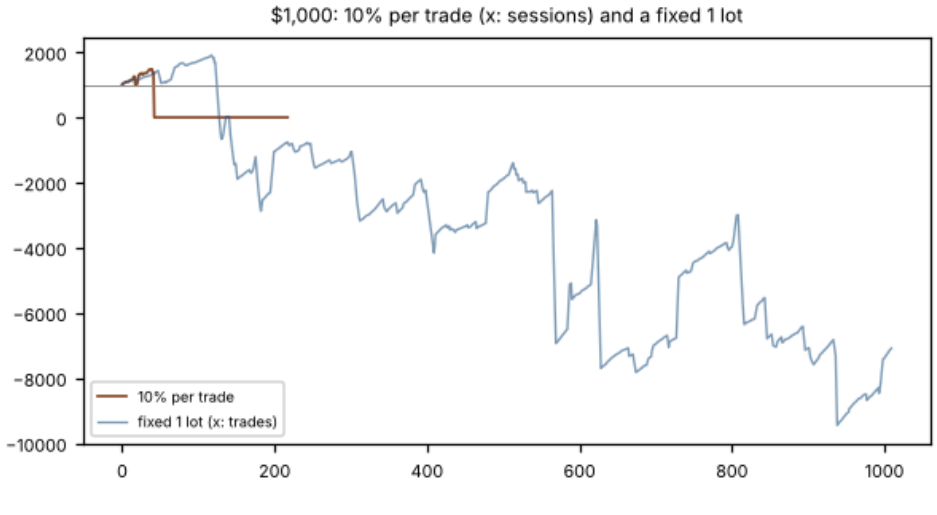
line (no verdict)	trades	mean	95 % interval	total $
sensitivity: EOD-marked sessions out	0	—	—	every in-test session carries an EOD mark (stage 1): n = 0

by year	2017	2018	2019	2020	2021	2022
trades	123	298	118	195	174	102
mean %	+3.01	-9.75	+5.77	-9.05	-8.46	+2.39
total $	$699	−$5,081	$1,141	−$2,602	−$2,741	$495

by weekday	Mon	Tue	Wed	Thu	Fri
trades	277	1	377	17	338
mean %	-8.63	+4.11	-14.82	+4.95	+8.99

entry hour	09:00	10:00	11:00	12:00	13:00	14:00	15:00
trades	6	76	118	104	145	233	328
mean %	+9.34	-11.31	-7.51	-1.99	-21.84	-4.35	+3.74





T5 — SPXW 13:00 ATM straddle to settlement, ex FOMC/CPI/NFP (T-E, H13)

Verdict: FAIL (mean > 0, Bonferroni lower end > 0, top-5 ≤ 100 %, ≥ 300 trades, ≤ 1 losing year). Unit: % of credit. Settlement: Cboe close (cash).

trades	sessions traded / in test	win %	avg win %	avg loss %	median %	mean %	mean $	total $	profit factor	max drawdown $	longest drawdown (sessions)
246	246 / 297	58.9	+54.78	-81.48	+17.09	**-1.16**	-57.60	−$14,170	0.83	$21,752	194

Best trade $2,172 (+86.5 %, 2020-04-15 13:00, SPXW straddle); worst −$7,318 (-567.3 %, 2018-02-05 13:00, SPXW straddle).

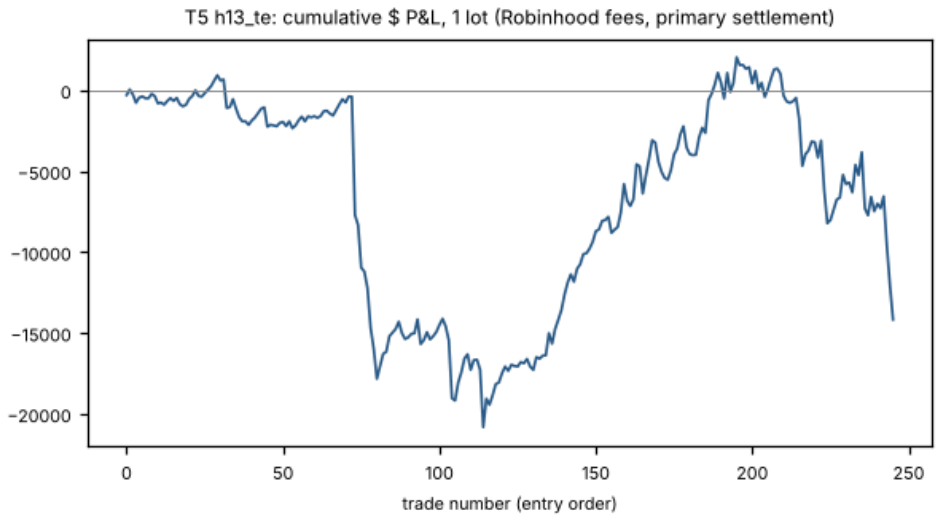
$1,000: at a fixed 1 lot it ends at −$13,170 (low −$19,819, first at or below $0 on 2017-04-05); at 10 % per trade (the final report's method B) it ends at $238. A short straddle or strangle can lose many times its credit, so 10 % of the balance on the credit can lose far more than 10 %.

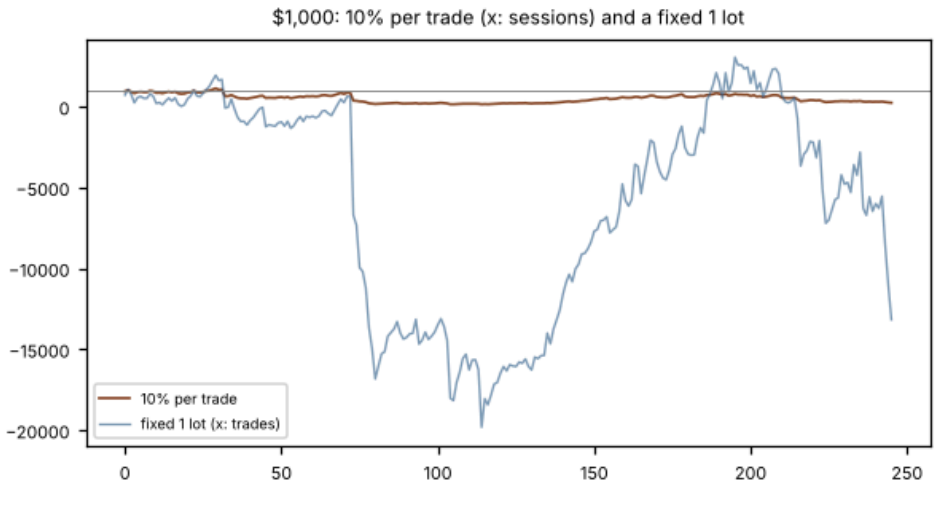
line (no verdict)	trades	mean	95 % interval	total $
primary, Robinhood fees	246	**-1.16**	[-12.26, +12.14]	−$14,170
primary settlement at the study's own fees	246	-1.12	[-12.22, +12.19]	−$14,107
sensitivity: Compass Close's P&L-sample sessions out	52	-6.83	[-22.98, +7.42]	−$2,207
sensitivity: EOD-marked sessions out	0	—	—	every in-test session carries an EOD mark (stage 1): n = 0

by year	2017	2018	2019	2020	2021	2022
trades	73	43	39	40	38	13
mean %	+1.67	-28.42	+33.57	+10.02	-16.91	-19.48
total $	−$386	−$18,671	$11,244	$8,289	−$6,779	−$7,866

by weekday	Mon	Tue	Wed	Thu	Fri
trades	93	9	81	4	59
mean %	+0.52	-27.72	-16.13	-27.08	+22.55

by entry time	13:00
trades	246
mean %	-1.16





T6 — SPXW 14:05 strangle at ±0.5 EM to settlement, ex FOMC and month-end (H15)

Verdict: FAIL (Bonferroni lower end > 0, top-5 ≤ 100 %, ≥ 300 trades, ≤ 1 losing year). Unit: % of credit. Settlement: Cboe close (cash).

trades	sessions traded / in test	win %	avg win %	avg loss %	median %	mean %	mean $	total $	profit factor	max drawdown $	longest drawdown (sessions)
280	280 / 297	72.9	+90.53	-234.26	+97.23	**+2.37**	-7.68	−$2,151	0.96	$14,425	161

Best trade $1,548 (+99.9 %, 2020-04-09 14:05, SPXW strangle); worst −$6,078 (-1146.8 %, 2018-02-05 14:05, SPXW strangle).

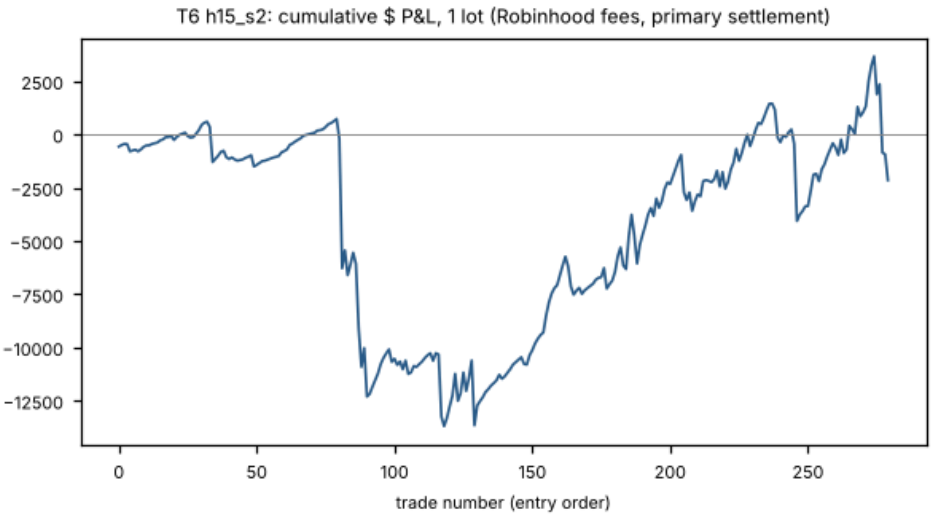
$1,000: at a fixed 1 lot it ends at −$1,151 (low −$12,695, first at or below $0 on 2017-04-05); at 10 % per trade (the final report's method B) it ends at $0, wiped out on 2017-04-05. A short straddle or strangle can lose many times its credit, so 10 % of the balance on the credit can lose far more than 10 %.

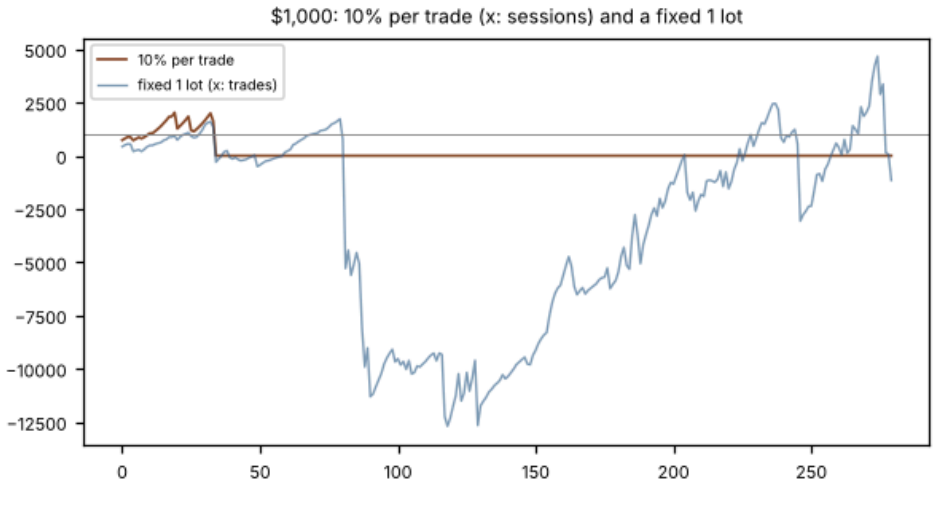
line (no verdict)	trades	mean	95 % interval	total $
primary, Robinhood fees	280	**+2.37**	[-27.19, +30.73]	−$2,151
primary settlement at the study's own fees	280	+2.45	[-27.11, +30.82]	−$2,098
sensitivity: Compass Close's P&L-sample sessions out	55	-18.36	[-129.79, +41.82]	−$1,182
sensitivity: EOD-marked sessions out	0	—	—	every in-test session carries an EOD mark (stage 1): n = 0

by year	2017	2018	2019	2020	2021	2022
trades	80	51	45	46	43	15
mean %	+15.51	-60.84	+53.84	+3.82	-2.07	+1.10
total $	$730	−$13,462	$6,033	$4,484	$1,511	−$1,447

by weekday	Mon	Tue	Wed	Thu	Fri
trades	92	7	90	4	87
mean %	+3.40	-16.95	-42.09	+11.64	+48.40

by entry time	14:05
trades	280
mean %	+2.37





T7 — QQQ overnight reversal of the late-day move (Compass H1)

Verdict: FAIL (Bonferroni lower end > 0, ≥ 300 trades). Unit: % of notional. Settlement: stock, next 09:30 open.

trades	sessions traded / in test	win %	avg win %	avg loss %	median %	mean %	mean $	total $	profit factor	max drawdown $	longest drawdown (sessions)
249	249 / 259	58.6	+0.619	-0.678	+0.133	**+0.082**	+54.21	$13,497	1.29	$4,543	65

Best trade $2,219 (+3.0 %, 2026-06-22 16:00, QQQ short); worst −$2,014 (-3.4 %, 2026-04-07 16:00, QQQ short).

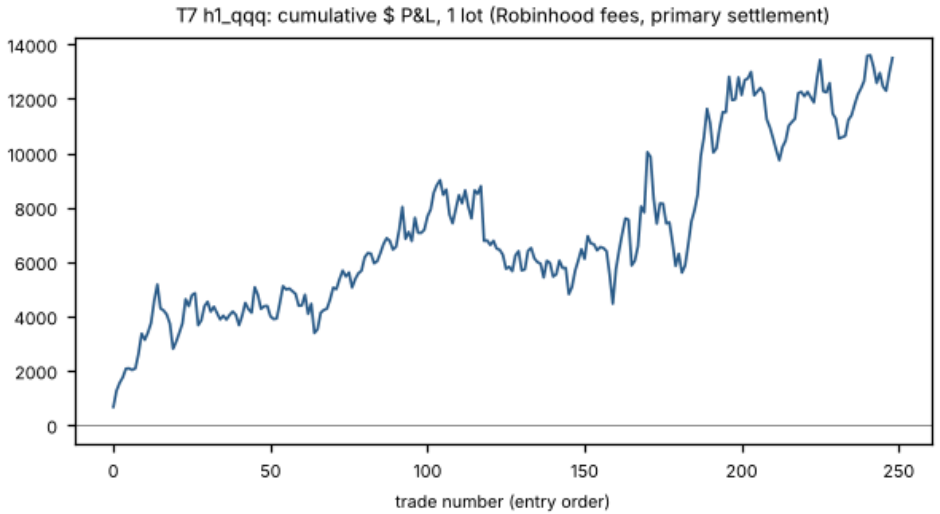
$1,000: at a fixed 1 lot it ends at $14,497 (low $1,000); at 10 % per trade (the final report's method B) it ends at $1,021.

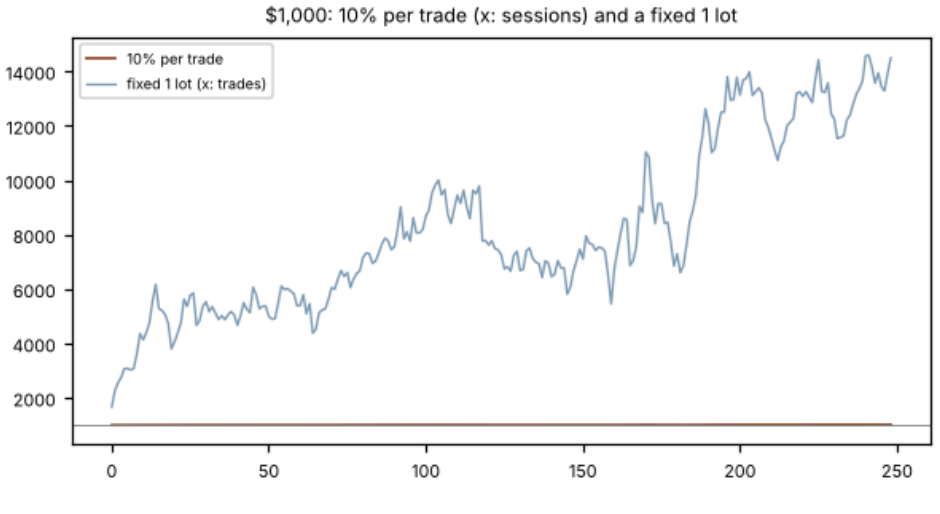
line (no verdict)	trades	mean	95 % interval	total $
primary, Robinhood fees	249	**+0.082**	[-0.007, +0.141]	$13,497
primary settlement at the study's own fees	249	+0.081	[-0.008, +0.140]	$13,345

by year	2025	2026
trades	54	195
mean %	+0.137	+0.067
total $	$4,493	$9,004

by weekday	Mon	Tue	Wed	Thu	Fri
trades	49	52	51	49	48
mean %	+0.166	+0.040	+0.073	+0.051	+0.086

by entry time	16:00
trades	249
mean %	+0.082





Integrity

- **Plain-loop re-derivation** (`verify_plain.py`): 70 trades, 10 a test (seed 20261024), re-derived from the raw files with separate code (pyarrow rows into Python lists; no polars, numpy, run script, engine or fee module): strikes, fills, max risk, C, D, O, the Cboe close, the case (b) and stock P&L at Robinhood's fees. **490 fields, 0 differences** (tolerance 1e-6).
- **Lookahead check** (`lookahead.py`, 10 sessions a family, seed 20261024; every chain row and bar stamped after the entry minute blanked, the settlement inputs kept; control: the chain read one minute later):

test	entries checked	sessions	unchanged with the future blanked	control changed
all	—	—	**not run — usage ended**	—

- **Units asserted** in the run code (prices per share, $ = 100 × points − fees, % = $ ÷ the unit's $ base); the SPXW P&L is re-asserted against its legs for every trade (`run_t56.py`).
- **Every run path reproduced the committed research rows** on previously read sessions before it touched fresh data (FETCH_LOG.md): the replay 110 of 110 trades, the touch events and hyp B rows exactly, T-E/H15 rows at both fees, H1 21 of 21.
- **Frozen code:** sim.py and load.py from `7dfea57`, sha256 in `out/t56_code_sha256.json`; the tracker at `rules-2026-10-21.1` / `rules-c3-2026-10-21.1`; the engine audit's and hyp B's engines unchanged; AFTERHOURS's `case_pnl` imported.

Deviations and disclosures

- **Four SDK logins in all:** two for the stage-1 entitlement probes (non-expiry days only, before the prereg), one for the fetch, one for the prereg's retry of the 8 empty QQQ answers (still empty, not filled in).
- **T7 ran before the fetch commit** (`f6d2889`): it needed no fetch, its code is the committed `run_t7.py` unchanged, and it ran once.
- **The lookahead check was not completed: usage ended.** It was started, restarted, and stopped before writing any result; no output was read. The remaining integrity work was skipped for the same reason. The plain-loop re-derivation (0 differences) and the reproduction of the committed rows on previously read sessions did run.
- **QQQ's third Fridays before 2015** are not in the test: the standard monthlies were then dated on the Saturday, and the frozen rules ask for the chain whose expiration is the session (prereg §2.2).
- **Every fresh session carries an end-of-day mark** from the Lifecycle daily tier or the Compass features (not counted as reads, as Compass Close decided), so the "EOD-marked sessions out" sensitivity line has no sessions.
- **The sensitivity line without Compass Close's P&L-sample sessions** leaves few SPY sessions (most fresh SPY sessions are in it); where it has fewer than 60 sessions it prints no interval.
- **T4's by-time table** is by entry hour (a touch can fall on any minute).
- **The study's own fees** are a reported line throughout; no verdict was read at them. Robinhood is cheaper on the ETF options and the stock trades and dearer on SPXW, so the mean at the study's fees is higher only for T5 (+0.04 points), T6 (+0.08 points); no verdict would differ.

Files

`research_notes/Compass Final/`: `PREREG_COMPASS_FINAL.md` and the stage-1 files (`4bc546f`); `fetch_final.py`, `FETCH_LOG.md` and the run code (`f6d2889`); the trade files `out/t123_trades.parquet`, `out/t4_trades.parquet`, `out/t56_trades.parquet` (+ `t56_skips.csv`), `out/t7_trades.parquet` (+ `t7_sessions.parquet`); `out/results.json`, `out/verify_plain.json`, `out/lookahead.json`, `out/t4_events.json`, `out/t123_replay_days.json`, `charts/`; this report and its PDF (`report_final.py`, `render_final.py`). S17's forward: `forward/s17_forward.py` (step 8 of `scripts/daily_forward.py`), `tests/test_s17_forward.py`. Raw data stay in the scratchpad (`$SP/compass_final/`).